pure rolling problems could be solved by considering the motion as any one of the following two cases.

(i) The combination of translational motion and rotational motion about the centre of mass. (or)

(ii) The momentary rotational motion about the point of contact.

EXAMPLE 5.22

A solid sphere is undergoing pure rolling. What is the ratio of its translational kinetic energy to rotational kinetic energy?

Solution

The expression for total kinetic energy in pure rolling is,

$$KE = KE_{TRANS} + KE_{ROT}$$

For any object the total kinetic energy as per equation 5.58 and 5.59 is,

$$KE = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}Mv_{CM}^2\left(\frac{K^2}{R^2}\right)$$

$$KE = \frac{1}{2}Mv_{CM}^2\left(1+\frac{K^2}{R^2}\right)$$

Then,

$$\frac{1}{2}Mv_{CM}^2\left(1+\frac{K^2}{R^2}\right) = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}Mv_{CM}^2\left(\frac{K^2}{R^2}\right)$$

The above equation suggests that in pure rolling the ratio of total kinetic energy, translational kinetic energy and rotational kinetic energy is given as,

$$KE : KE_{TRANS} : KE_{ROT} :: \left(1+\frac{K^2}{R^2}\right) : 1 : \left(\frac{K^2}{R^2}\right)$$

Now, $KE_{TRANS} : KE_{ROT} :: 1 : \left(\frac{K^2}{R^2}\right)$

For a solid sphere, $\frac{K^2}{R^2} = \frac{2}{5}$

Then, $KE_{TRANS} : KE_{ROT} :: 1 : \frac{2}{5}$ or

$$KE_{TRANS} : KE_{ROT} :: 5 : 2$$

5.6.4 Rolling on Inclined Plane

Let us assume a round object of mass m and radius R is rolling down an inclined plane without slipping as shown in Figure 5.37. There are two forces acting on the object along the inclined plane. One is the component of gravitational force (mg sinθ) and the other is the static frictional force (f). The other component of gravitation force (mg cosθ) is cancelled by the normal force (N) exerted by the plane. As the motion is happening along the incline, we shall write the equation for motion from the free body diagram (*FBD*) of the object.

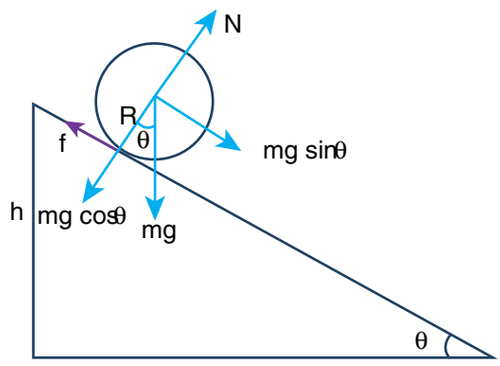


Figure 5.37 Rolling on inclined plane

UNIT-5(XI-Physics_Vol-1).indd 256 06-01-2022 17:48:10